

Nile University

Nanoelectronics Integrated Systems Center (NISC)

Task 3

Evaluating Chaotic Systems

Prepared by

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Part A: Phase Space Reconstruction (PSR)

To reconstruct the phase space of the Lorenz system from the scalar time series $x(t)$, we estimated the optimal time delay (τ) and embedding dimension (m).

- **Time Delay (τ):** Using the Average Mutual Information (AMI) method, the first local minimum was found at $\tau = 17$.
- **Embedding Dimension (m):** Using the False Nearest Neighbours (FNN) method, the percentage of false neighbours dropped below the 1% threshold at $m = 3$. This perfectly matches the known state dimension of the Lorenz system.

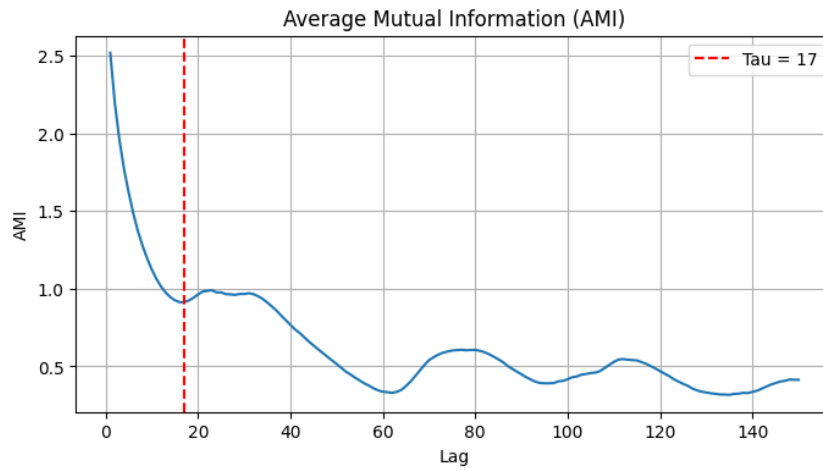


Figure 1: Average Mutual Information (AMI) vs. Lag, showing the selected delay $\tau = 17$.

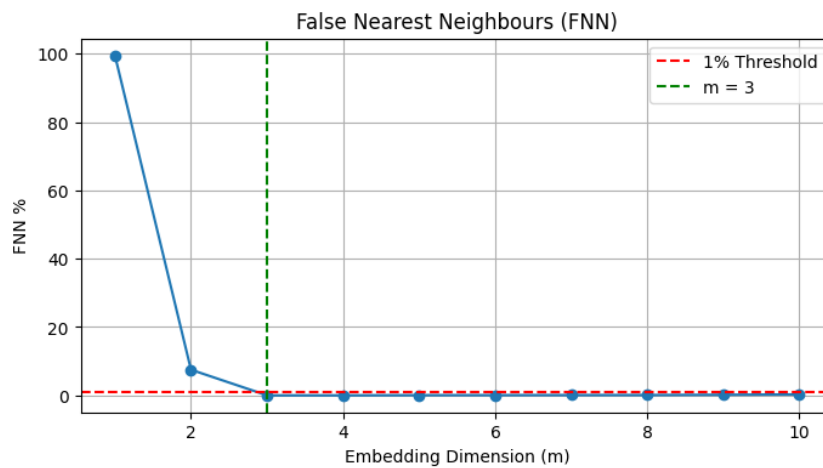


Figure 2: False Nearest Neighbours (FNN) percentage vs. Embedding Dimension, indicating $m = 3$.

Part B: Lyapunov Exponent

The Lyapunov exponents quantify the rate of separation of infinitesimally close trajectories. We computed the exponents using two different methods:

- **From the ODE (Wolf's Algorithm):** The full Lyapunov spectrum was computed as $[1.2772, 0.0114, -20.9937]$ bits/time. The largest Lyapunov exponent (LLE) is $\lambda_1 \approx 1.2772$ bits/time.
- **From Time Series:** Using the reconstructed phase space ($\tau = 17, m = 3$), the estimated LLE is $\lambda_1 \approx 2.2603$ bits/time.

Discussion: Both methods successfully identify a strictly positive largest Lyapunov exponent, which is the hallmark of chaotic behavior. The discrepancy in the exact values is expected; the ODE method is highly precise as it utilizes the exact Jacobian, whereas the time series estimation is subject to finite sample length, embedding artifacts, and noise.

Part C: Kolmogorov (K2) Entropy

Kolmogorov-Sinai (K_2) entropy estimates the rate of information production in the system. Based on the correlation sums, the estimated K_2 values for increasing embedding dimensions are:

- $m = 2 \rightarrow 3$: 87.72
- $m = 3 \rightarrow 4$: 55.79
- $m = 4 \rightarrow 5$: 50.71
- $m = 5 \rightarrow 6$: 47.02

Discussion: The K_2 entropy values are positive and demonstrate a clear tendency to saturate (converge) as the embedding dimension increases. This finite, positive saturation indicates that the signal originates from a deterministic chaotic system (low-dimensional chaos) rather than random noise, which would exhibit unboundedly growing entropy.

Part D: Correlation Dimension

The correlation dimension (D_2) was estimated using the Grassberger-Procaccia algorithm by analyzing the correlation sum $C(r)$. The D_2 values extracted from the slopes of the linear scaling regions for dimensions $m = 2$ to 6 are approximately: 1.65, 1.89, 1.97, 2.04, 2.07.

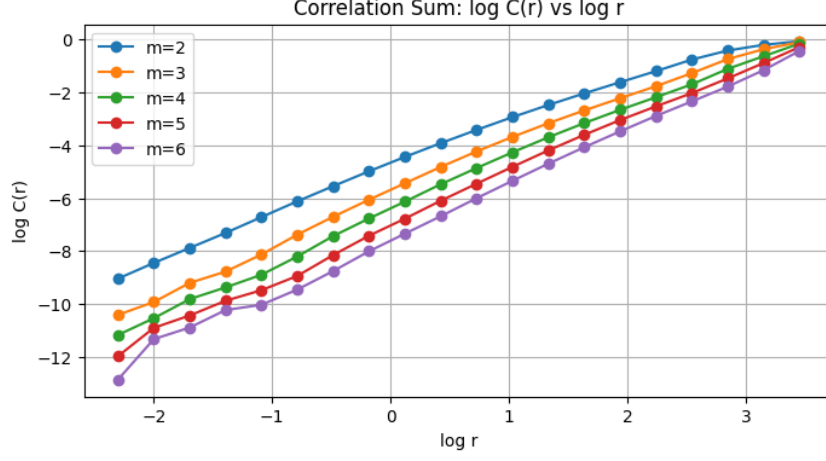


Figure 3: Correlation Sum: $\log C(r)$ vs. $\log r$ for embedding dimensions $m = 2$ to 6.

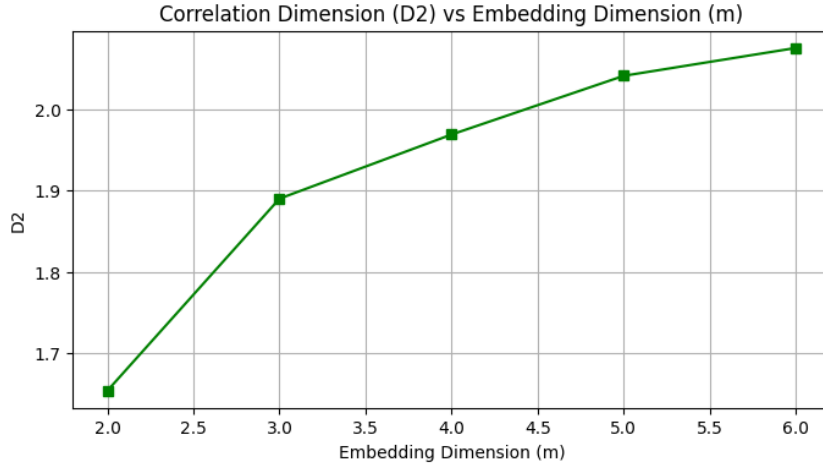


Figure 4: Correlation Dimension (D_2) vs. Embedding Dimension (m).

Discussion: As the embedding dimension m increases, D_2 saturates at a non-integer value of approximately **2.05**. This fractional dimension is a defining characteristic of a strange attractor.

Part E: Fractal (Box-Counting) Dimension

The Box-Counting dimension (D_0) offers an alternative geometric measure of the attractor's fractality.

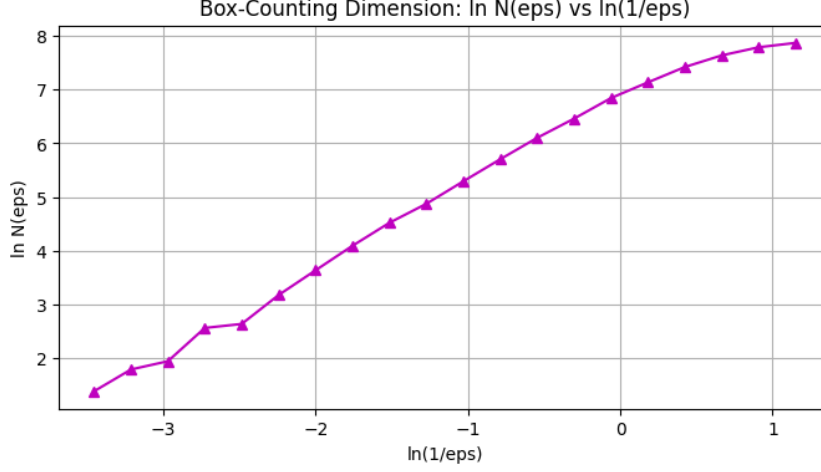


Figure 5: Box-Counting Dimension: $\ln N(\epsilon)$ vs. $\ln(1/\epsilon)$ showing the linear scaling region.

Discussion: The extracted slope from the scaling region yields a box-counting dimension of $D_0 \approx 1.5194$. While theoretically $D_0 \geq D_2$, practical numerical estimations from finite time series often yield approximations. Nonetheless, the non-integer value corroborates the fractal nature of the Lorenz attractor.

Conclusion

In conclusion, the quantitative evaluation confirms that the Lorenz system is deterministic and chaotic, but not hyperchaotic, as evidenced by the presence of exactly one positive Lyapunov exponent. Furthermore, the positive and saturating Kolmogorov (K_2) entropy indicates that the system loses predictability at a finite rate; this means its behavior can be forecasted in the short term, but long-term prediction is fundamentally impossible due to sensitive dependence on initial conditions. Finally, the non-integer values obtained for both the correlation dimension (D_2) and box-counting dimension (D_0) prove that the system's strange attractor is neither a simple two-dimensional surface nor a solid three-dimensional volume. Instead, it forms a complex fractional (fractal) structure. Together, these metrics conclusively classify the system as exhibiting low-dimensional chaos.